

Advanced Animal Movement Modelling Training Workshop

Space Use Estimation

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Dispelling misconceptions

- Commonly used terms such as "space use", "home range", and "utilization distribution (UD)" are not interchangeable
 - Space use: estimating the geographic area where an animal has been (or will be)
 - Home range: predictable areas of restricted space use that an animal repeatedly visits
 - Utilization distribution (UD): a probability density function that characterizes space use

Separating concepts from methods

- Concepts related to space use include the general time-period of interest
 - Such as areas used during tracking period vs over the long term
- Methods involve how space use is estimated based on the chosen conceptual perspective
 - These methods can involve UD or not

What does your space-use estimate mean?

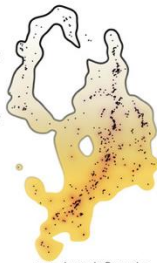
Concept: Occurrence distributions
Estimating the space potentially used by animals during the sampling period, essentially estimates of uncertainty between known locations.

Distribution: Occurrence distribution

Contour: Confidence area

Predictive scope: Within sample interpolation

Example question: What potential movement corridors does species A use?

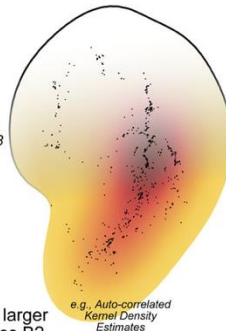


e.g. dynamic Brownian Bridge Movement Models

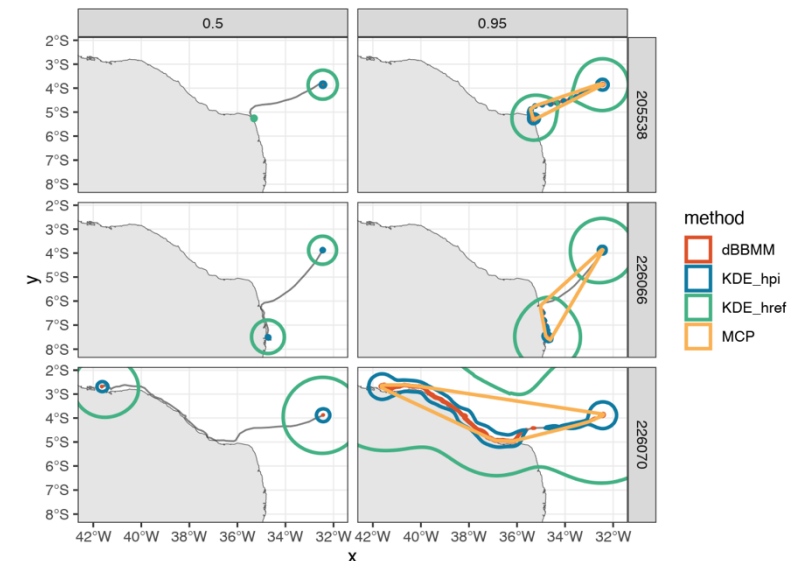
Range distributions
Estimating the space requirements of animals, corresponding to Burt's 1943 concept of home range.

Utilisation distribution
Home range
Beyond sample extrapolation

Does species A have a larger home range than species B?



e.g. Auto-correlated Kernel Density Estimates



Concepts

Concepts of space use

What does your space-use estimate mean?

Concept: Occurrence distributions

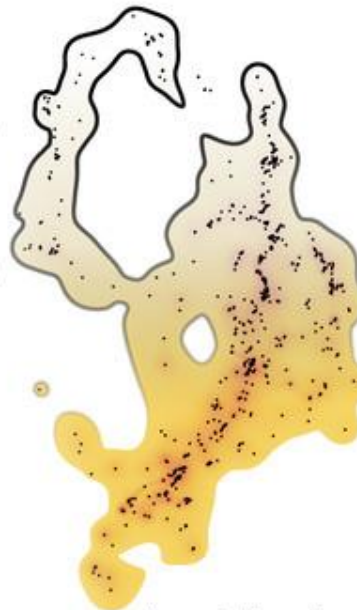
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*e.g., dynamic Brownian
Bridge Movement
Models*

Range distributions

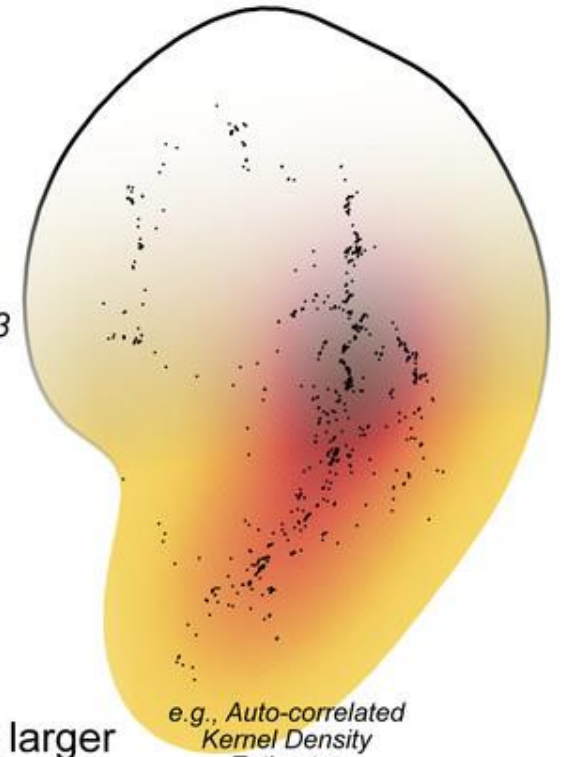
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Utilisation distribution

Home range

Beyond sample extrapolation

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*e.g., Auto-correlated
Kernel Density
Estimates*

Range distributions

Home range definition per Burt (1943):

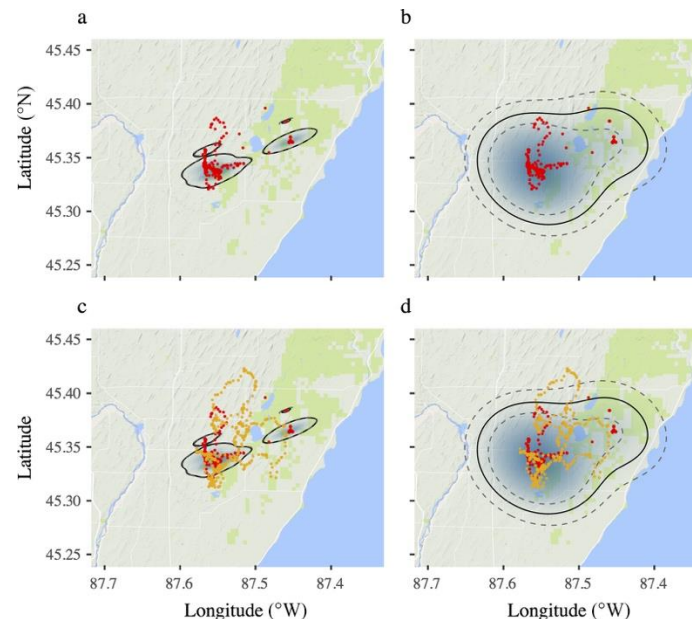
- “...that area traversed by the individual in its normal activities of food gathering, mating, and caring for young. Occasional sallies outside the area, perhaps exploratory in nature, should not be considered as part of the home range.”

Per Alston et al. (2026):

- “A range distribution describes the probability of an animal being in a location at a given time, taking into account all of the locations in a movement track simultaneously.”
- “Range distributions ... describe the long-run behavior of a movement process that features restricted space use and are consistent with Burt's classic definition of the home range.”

Range distributions

- Importantly, range distributions generate a predictive probability of space use as long as the underlying movement process does not change (Alston et al., 2026)
 - So these estimates aren't particularly well-handled for tracks that include migratory or dispersal components
 - Range distributions can probabilistically predict future space use



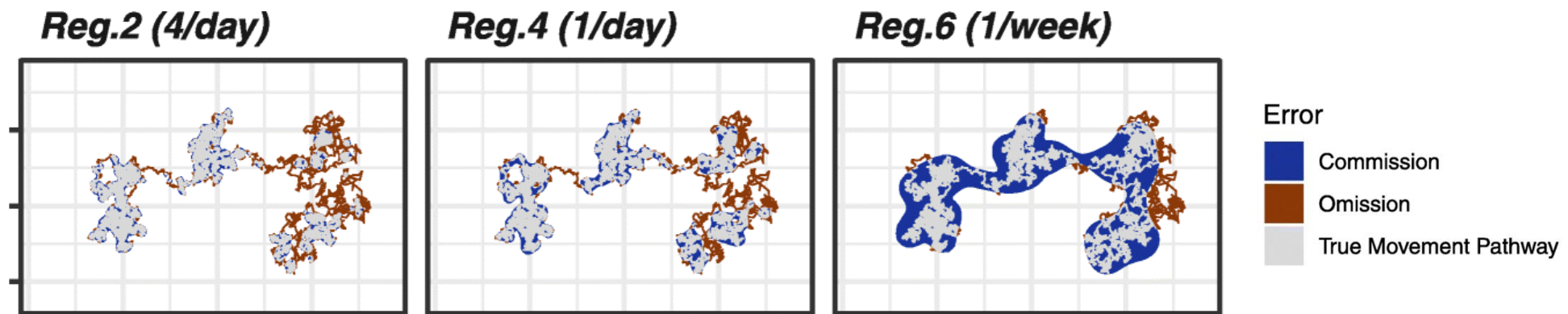
Occurrence distributions

Per Alston et al. (2026):

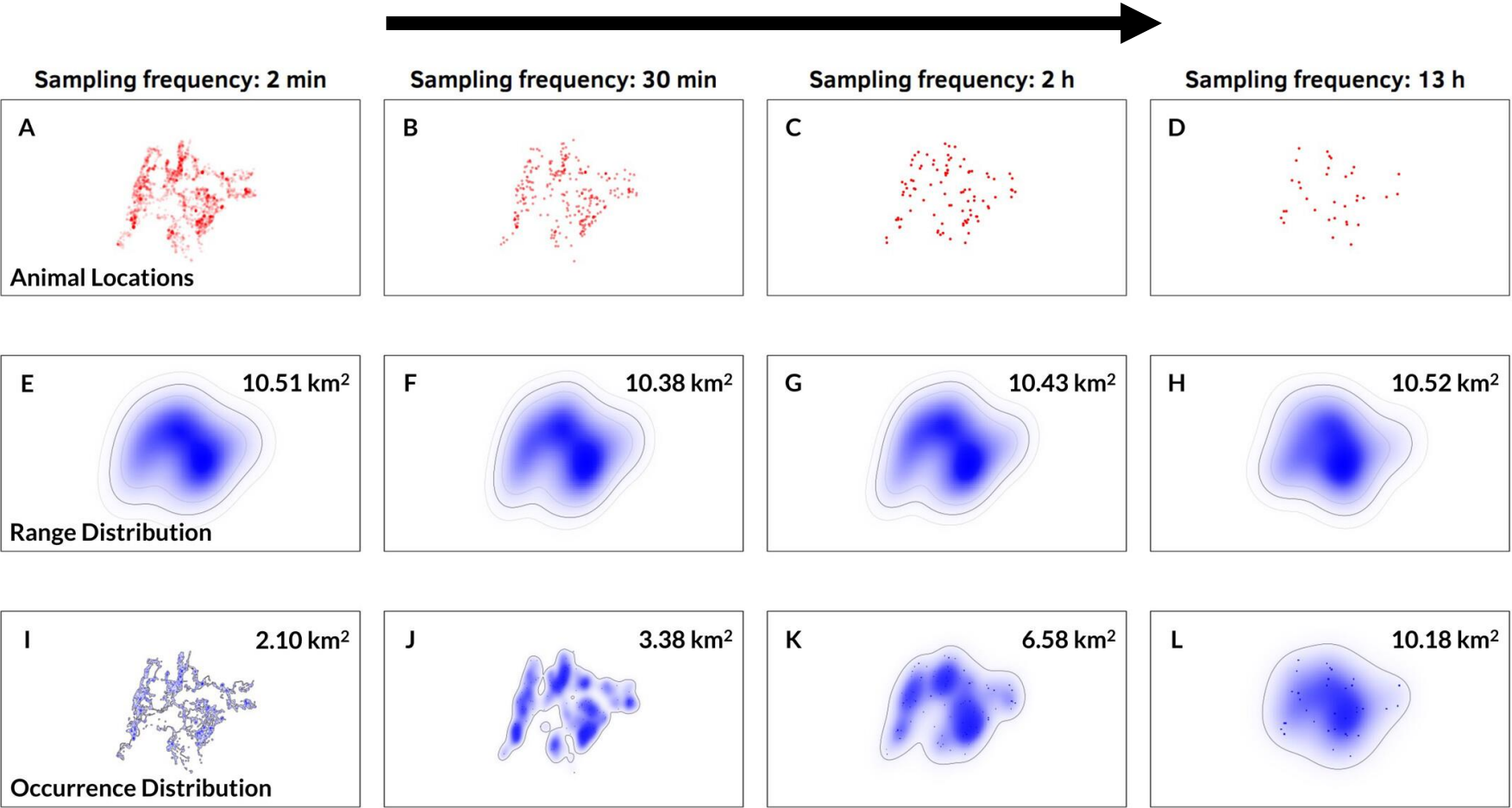
- “Occurrence distributions ... quantify uncertainty in the movement path of an individual during a period of observation and are not directly related to Burt's definition of the home range.”
- “...focused on interpolating the specific movement path that an animal took during an observation period and that connected a series of sampled locations.”
- “...there must be detectable autocorrelation in the movement process for estimators of occurrence distributions to produce informative predictions of movement between observed locations”

Occurrence distributions

- Importantly, occurrence distributions provide “confidence intervals” on animal trajectories but cannot predict them (Alston et al., 2026)
 - Technically doesn't measure space use
 - As sampling interval decreases (such as on the scale of seconds), the estimated area of the occurrence distribution would approach zero



Comparison by sampling rate



Example questions per concept

Range Distribution	Occurrence Distribution
How large is an animal's home range?	How likely is it that an animal visited a location of interest?
How do environmental and social factors influence animal area requirements?	Where might an animal have crossed a linear feature?
Where are individual animals likely to interact over time?	When and where could two individual animals have interacted?
What area is available to an animal in studies of third-order habitat selection?	In which areas of a landscape did an animal visit high-priority resources (e.g., migratory corridors, stopover sites, or feeding grounds)?

Methods

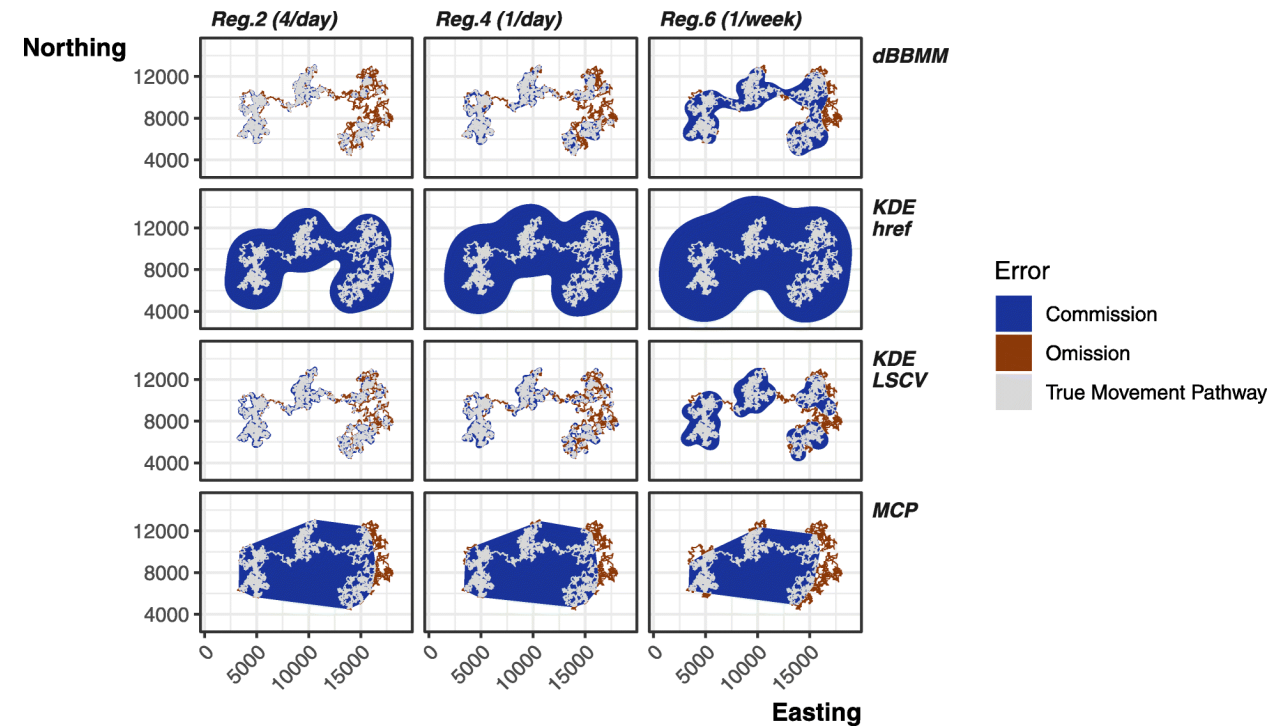
Common methods by concept

Range distribution

- Minimum convex polygon (MCP)
- Kernel density estimation (KDE)
 - Some versions
- Autocorrelated KDE (AKDE)

Occurrence distribution

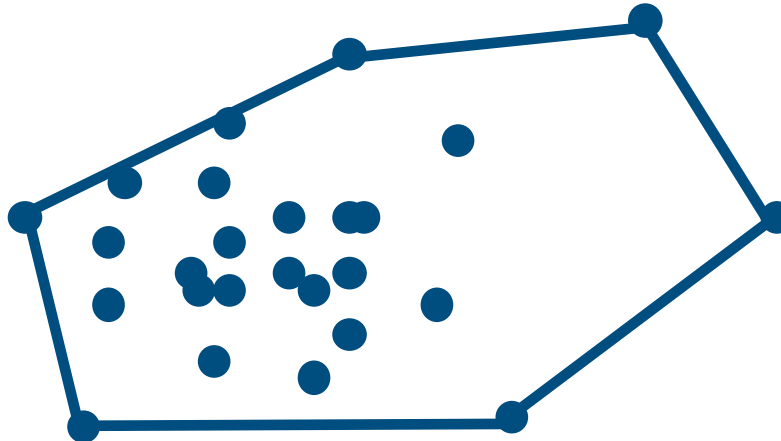
- Dynamic Brownian Bridge Movement Model (dBBMM)
- Kriging
- KDE (some versions)



Silva et al. 2020

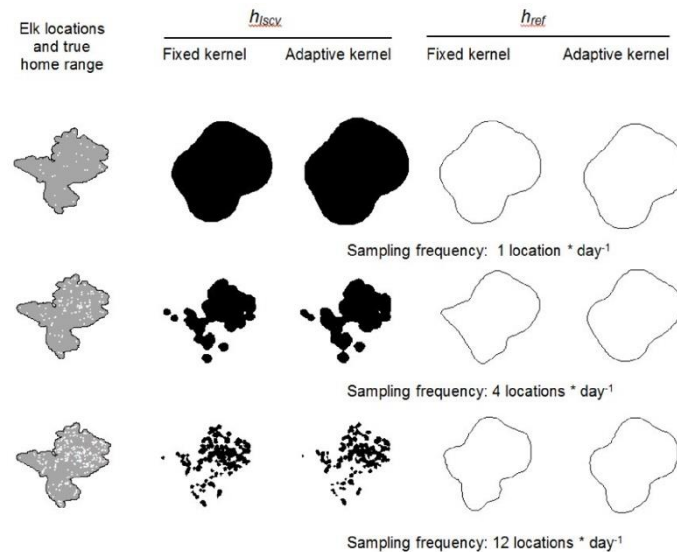
Minimum convex polygons (MCPs)

- Arguably the simplest method
- Create a convex hull that connects the outermost points
 - Extremely sensitive to outliers and sample size
- Doesn't account for spatial intensity of points

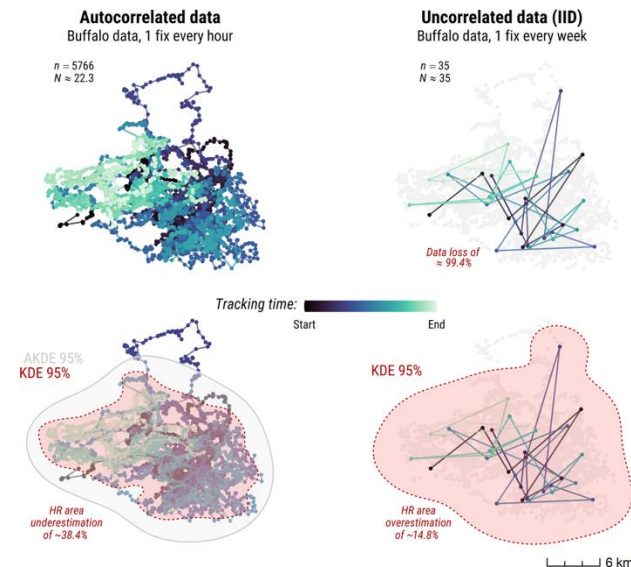


Kernel density estimation (KDE)

- Assumes that points are independent and identically distributed (IID)
- Bandwidth smoothing estimator (H) dictates how tight predicted surface is to points
 - Most common include “reference”, “least squares cross-validation”, and “plug-in” estimators
- Underestimates highly autocorrelated data and overestimates IID data



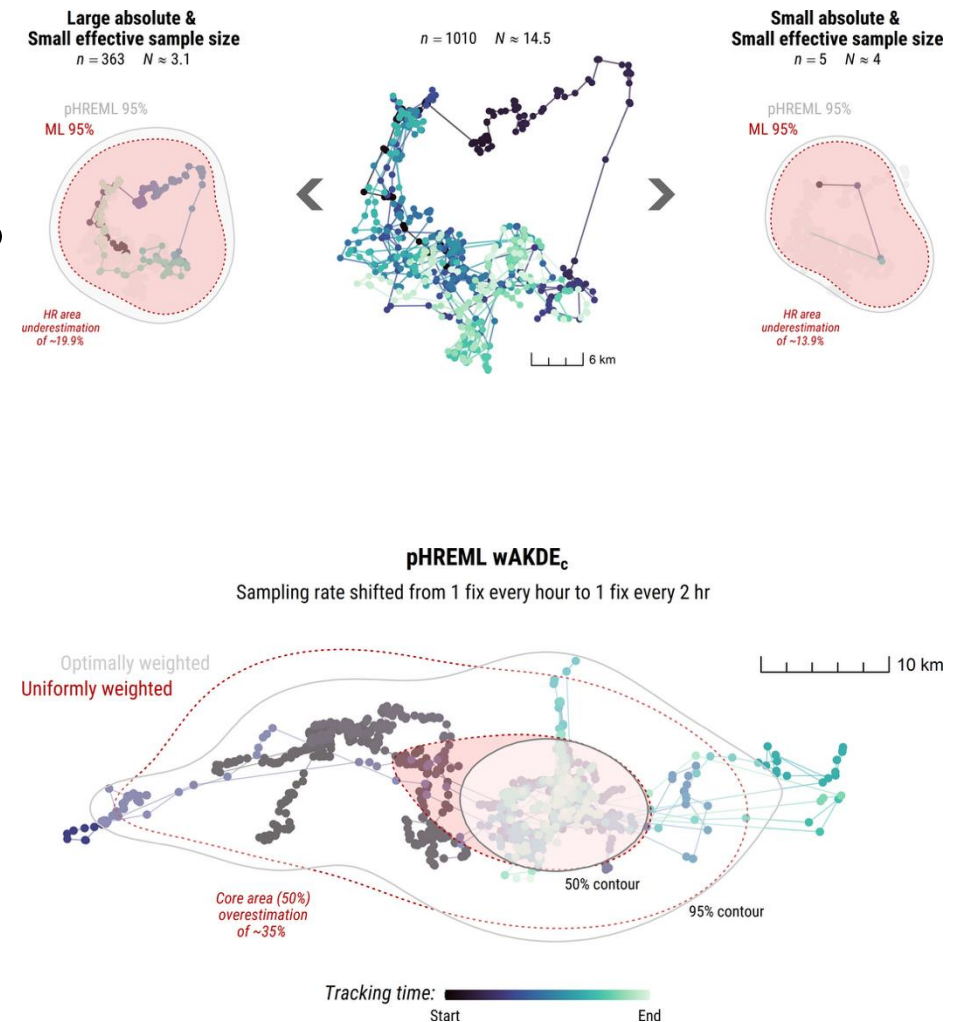
Kie 2013



Silva et al. 2022

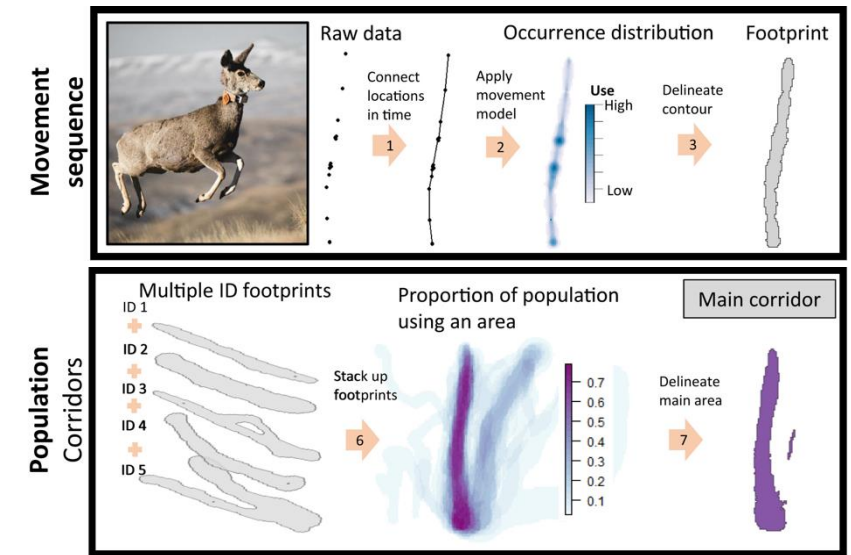
Autocorrelated KDE (AKDE)

- Explicitly requires a movement model that accounts for the autocorrelation in the tracking data
- Tests range of continuous-time movement models to find best fit to data
- Includes methods for bias correction related to small absolute and/or effective N
- Can accommodate irregular sampling intervals via weighted AKDE



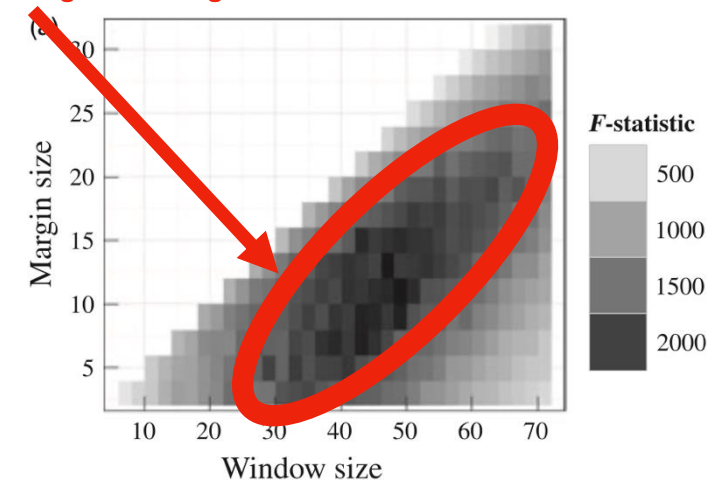
Dynamic Brownian bridge movement model (dBBMM)

- Extension to estimate Brownian motion variance (σ_m^2) for different behavioral segments of tracks (Kranstauber et al. 2012)
- Requires defining two parameters:
 - Margin size (m)
 - Must be minimum of 3
 - On either end of w
 - Sliding window size (w)
 - Must be $> 2m$
- Choices for w and m should be determined by the time interval over which relevant behavioral changes are expected
- Values will vary depending on the sampling interval



Merkle et al. 2022

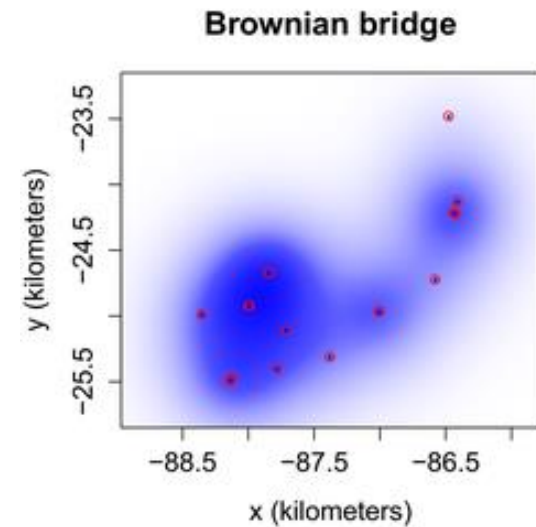
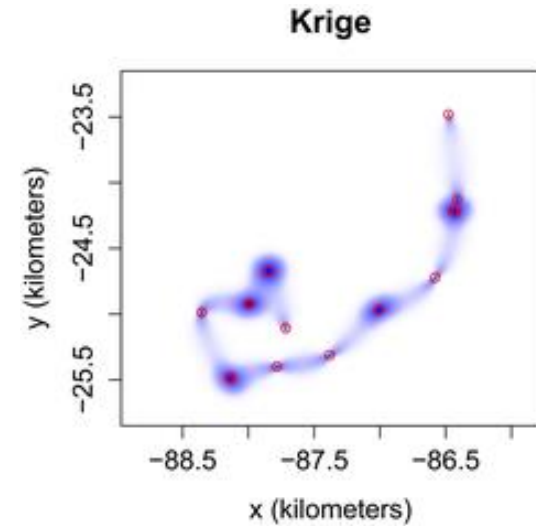
Values providing best fitting model



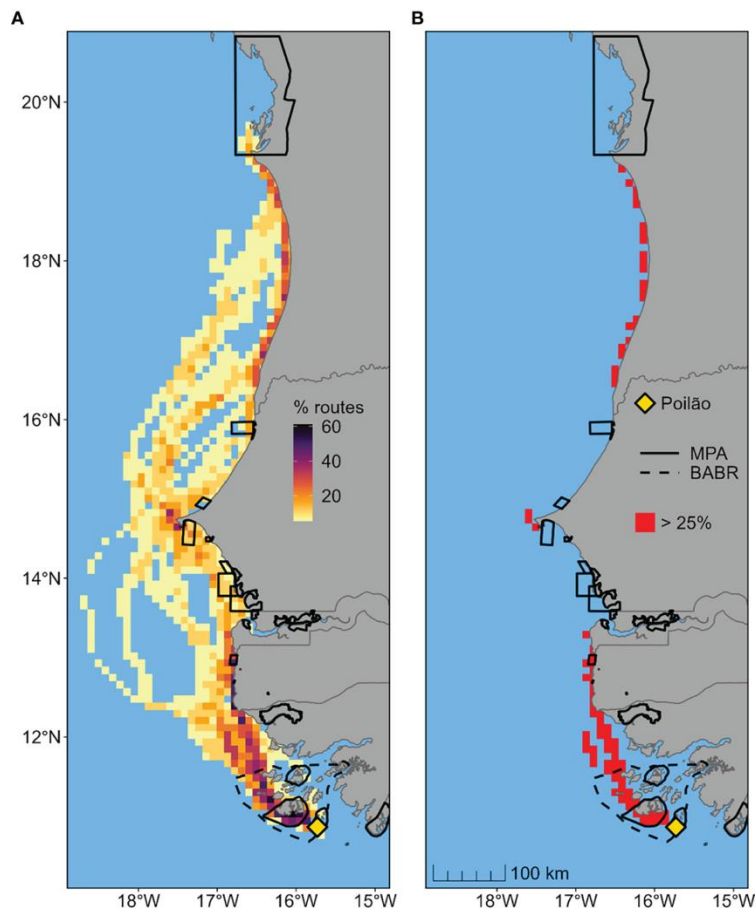
Kranstauber et al. 2012

Kriging

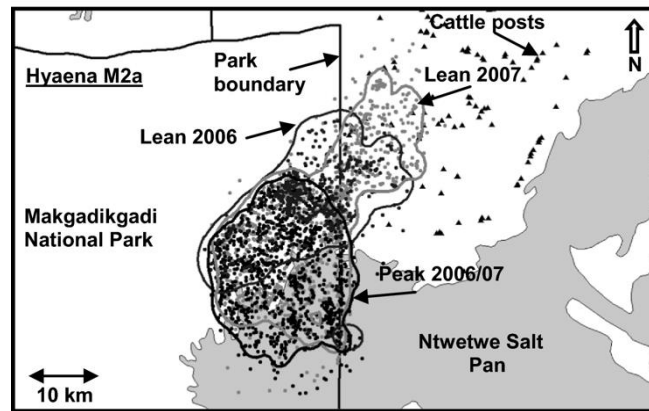
- Doesn't assume movement process (as dBBMM does)
 - Therefore also much quicker to fit
- Provides a statistically optimal method of model-based interpolation (Fleming et al. 2016)
- Represents general case of BBMM



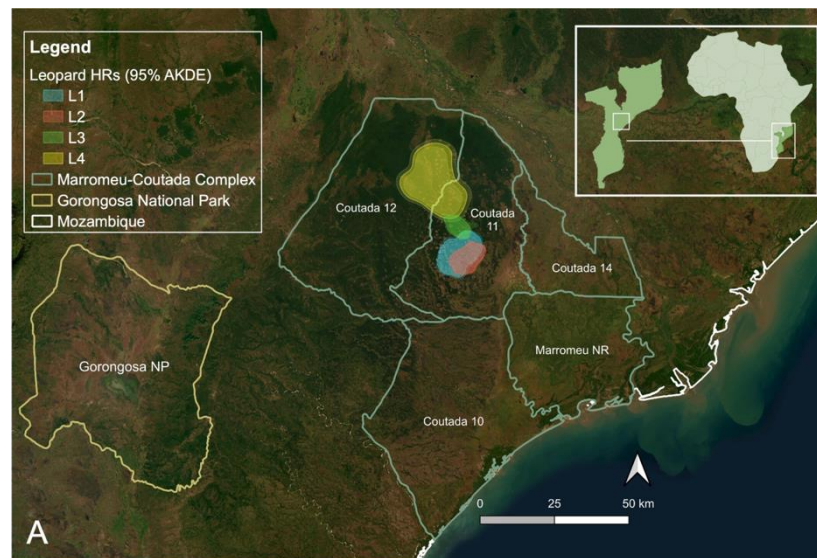
Real-world conservation applications



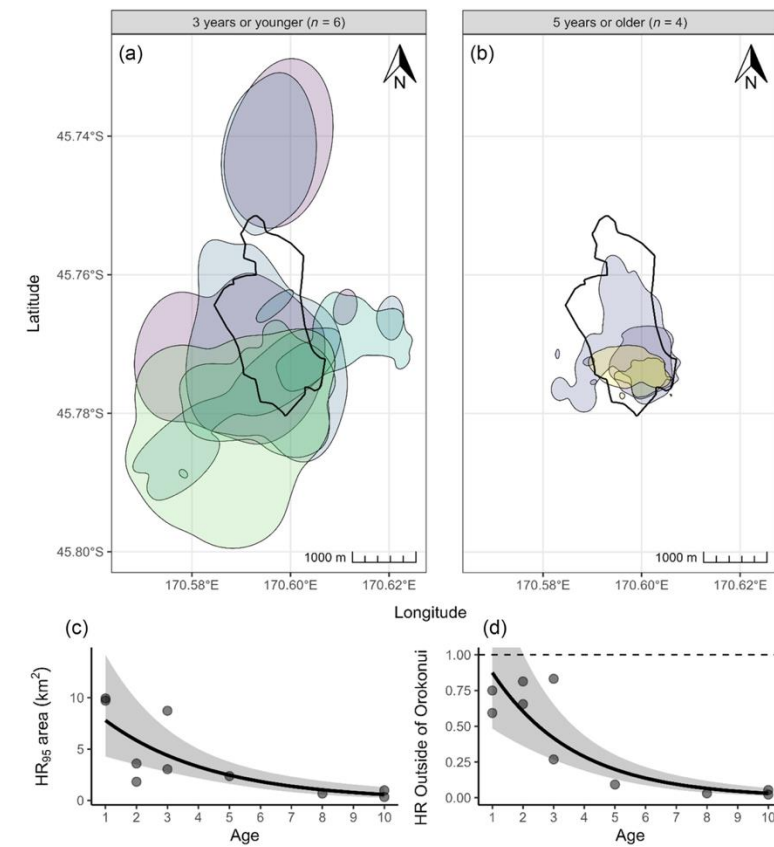
Patrício et al. 2022



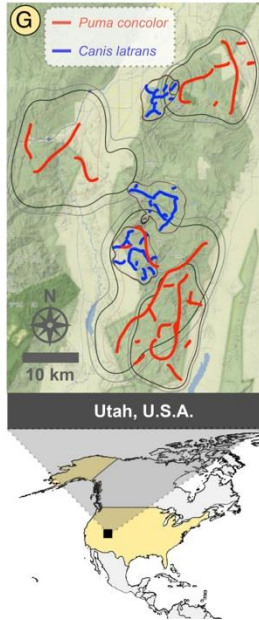
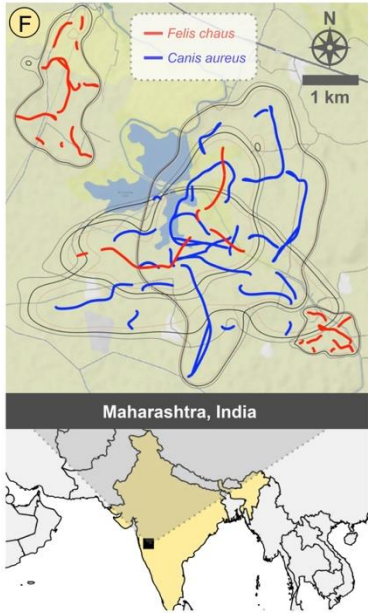
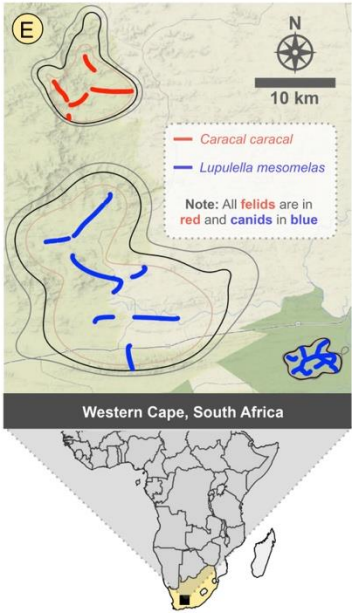
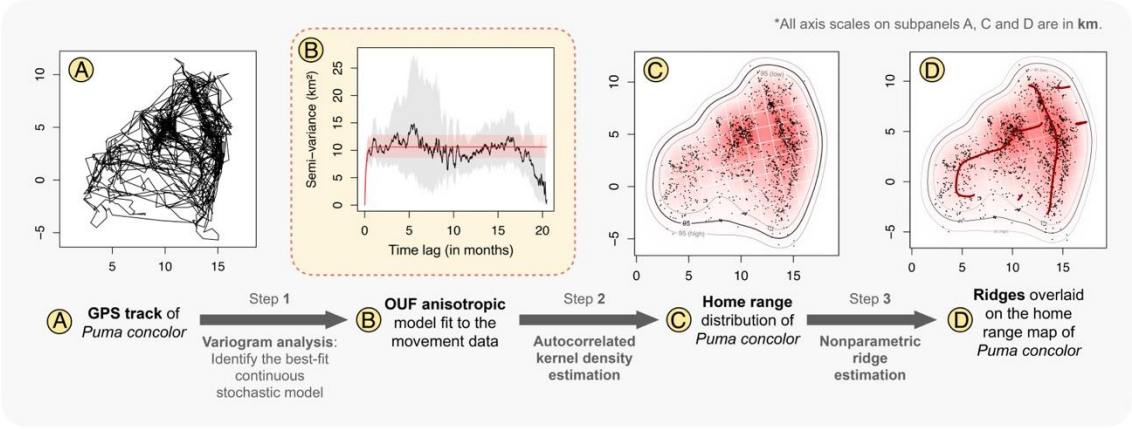
Maude et al. 2019



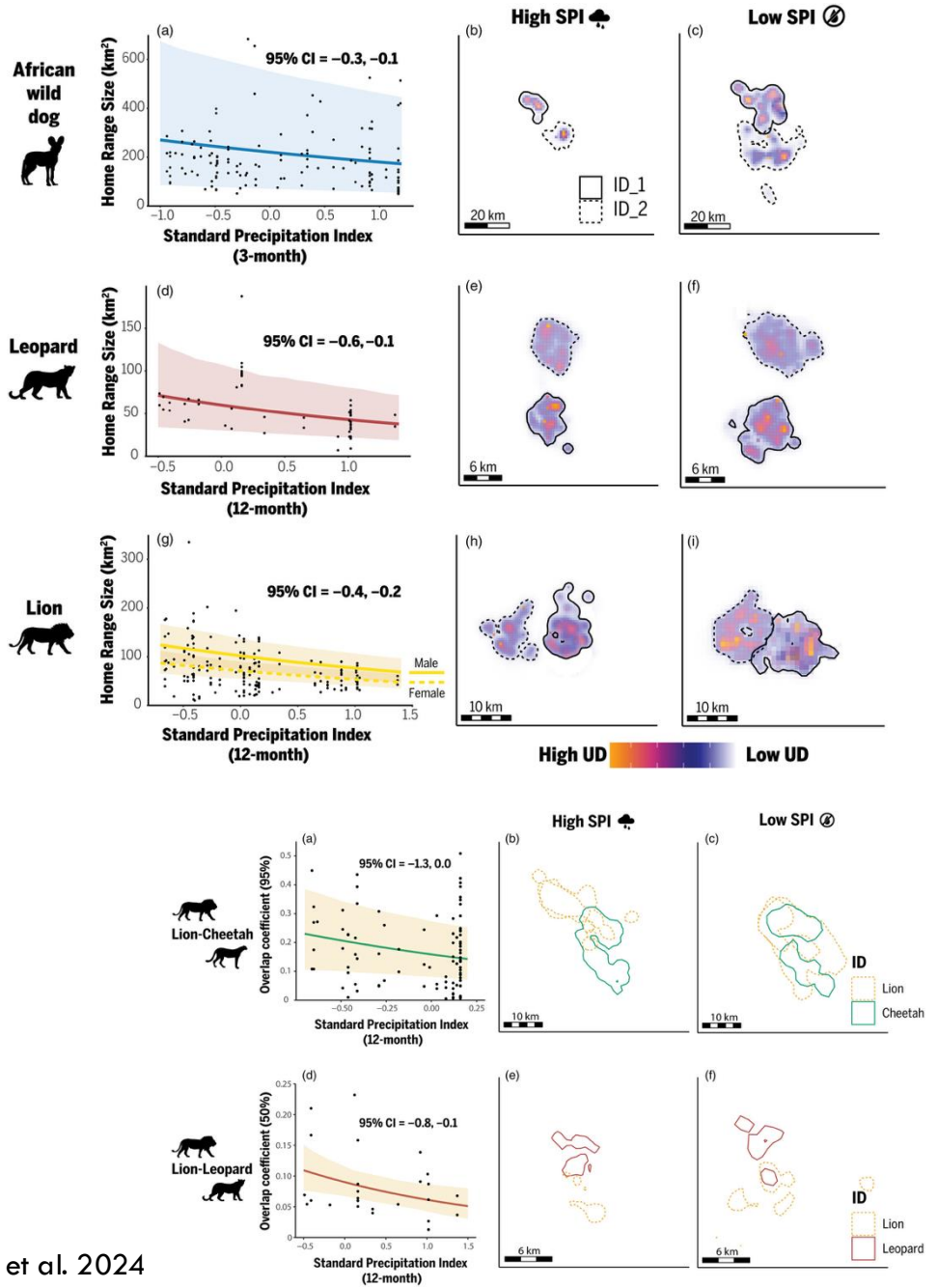
Briers-Louw et al. 2025



Forrest et al. 2024



Fagan et al. 2025



West et al. 2024

Time for live-coding



Estimate occurrence and range distributions with KDE,
dBBMM, and AKDE